

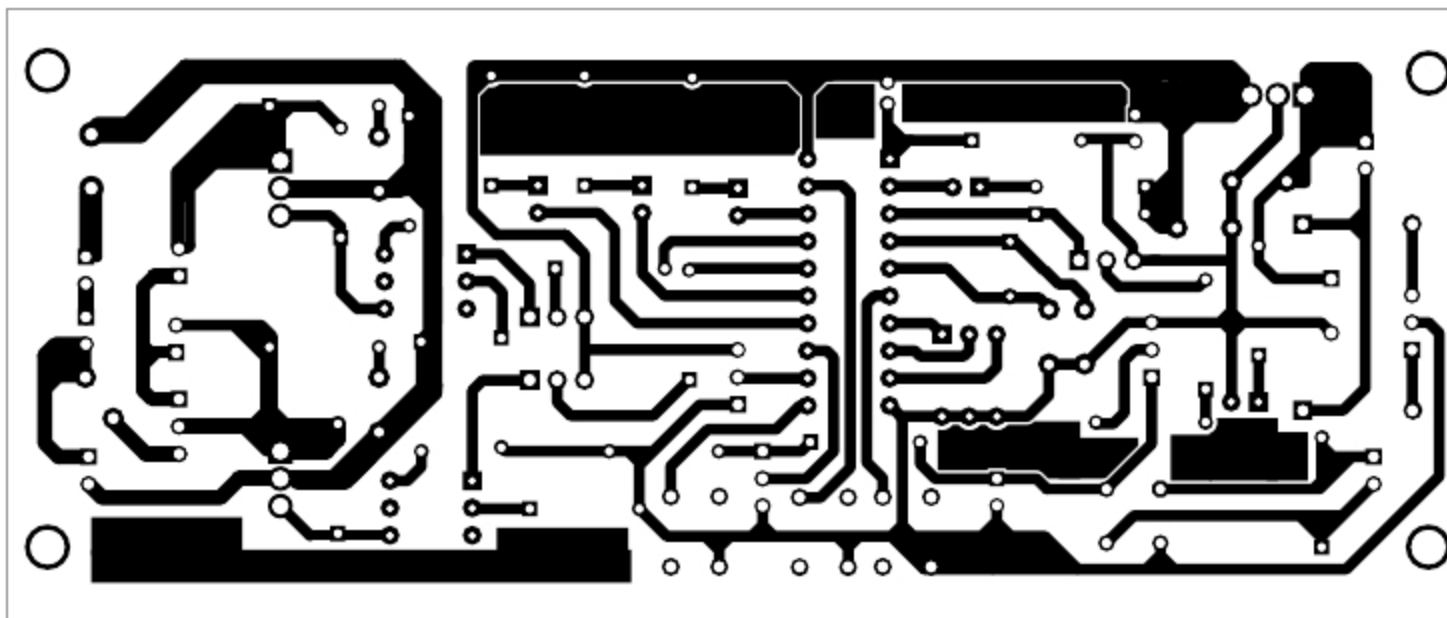


Fig. 3: Actual-size PCB layout of submersible motor control panel

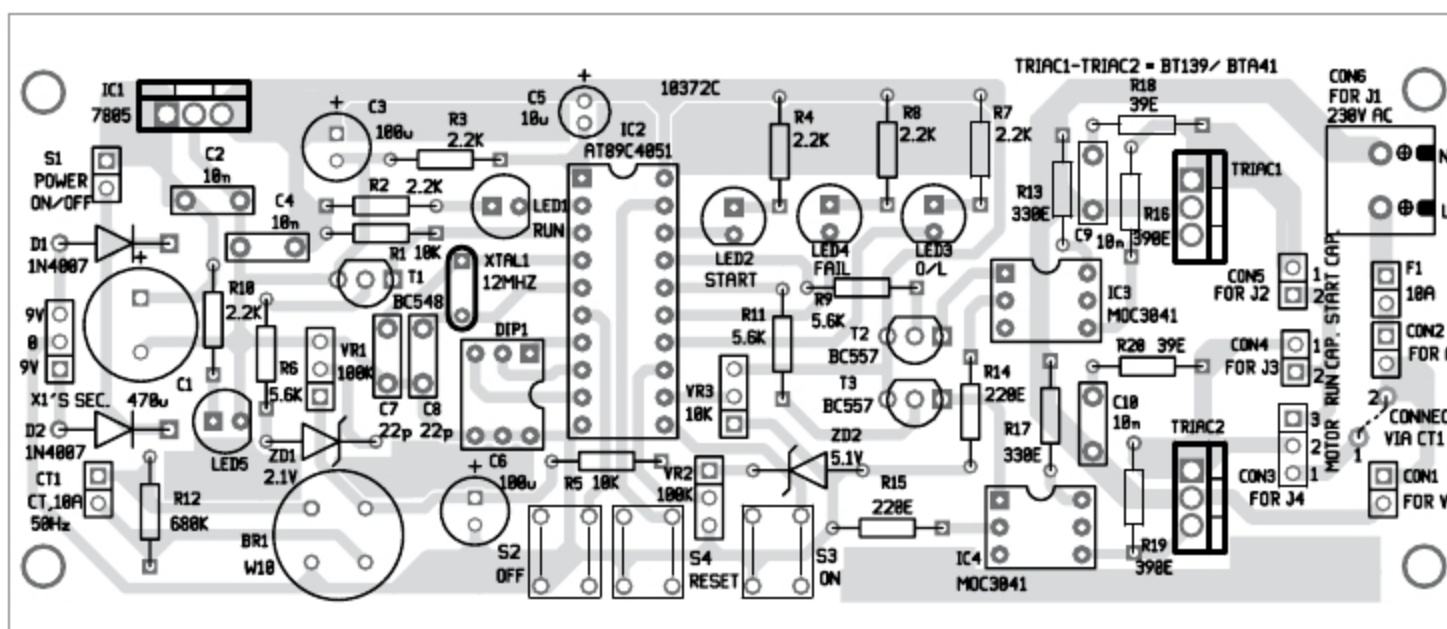


Fig. 4: Components layout for the PCB

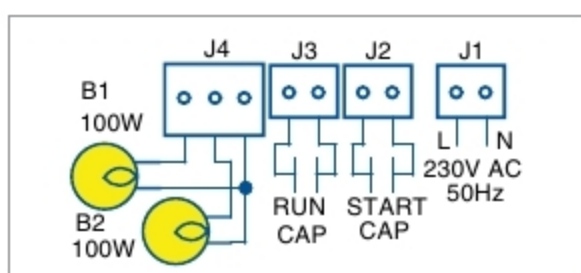


Fig. 5: Wiring diagram for testing

Reset switch (S4) can be used to break error loops of the software.

Construction and testing

An actual-size PCB layout of the submersible pump control panel is shown in Fig. 3 and its components layout in Fig. 4.

After assembling the circuit on the PCB, burn the hex code into the MCU AT89C4051 and insert it into the socket, connect 230V across CON6 and the primary of X1. Solder the coil of CT1 on the PCB and pass the mains live (L) wire inside CT1 after fuse via points 1 and 2, as shown in the PCB.

and LED1 goes off. Ensure that both lamps connected at J4 are off. Turn VR2 in one direction and press S4 momentarily. Press S4, and LED1 will light up. Both lamps connected at J4 will also light up at almost the same brightness. After three seconds, one lamp connected to J4 will become dim.

Ensure that the system stays in this condition for more than two minutes. If not, repeat the same process by varying VR2 as in last rotated direction.

Rotate VR1 fully in clockwise direction. Wait for one minute for LED3 to light up, and LED1 and LED2 to go off. If that does not happen, rotate VR1 fully in anticlockwise direction, and repeat the same process. Rotate VR1 slowly in reverse direction and

ensure that the system stays in on condition for more than two minutes.

In brief, at normal condition, keep around 0.45V at pin 12 of IC2 and 0.9V at pin 13 of IC2, using VR3 and VR2, respectively. Also, when normal current is flowing through CT1, pin 3 of IC2 remains low. This becomes high when higher current flows through CT1.

After proper calibration, fix the entire assembly in a suitable enclosure and connect the motor after removing the bulbs.

Switch on the unit and ensure that water is flowing from the pump, and current and voltage indications are normal. Adjust VR2 slowly to hold the system in normal condition without an overload trip. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



A. Asokan Ambali is working as senior instrument mechanic at Naval Aircraft Yard Kochi